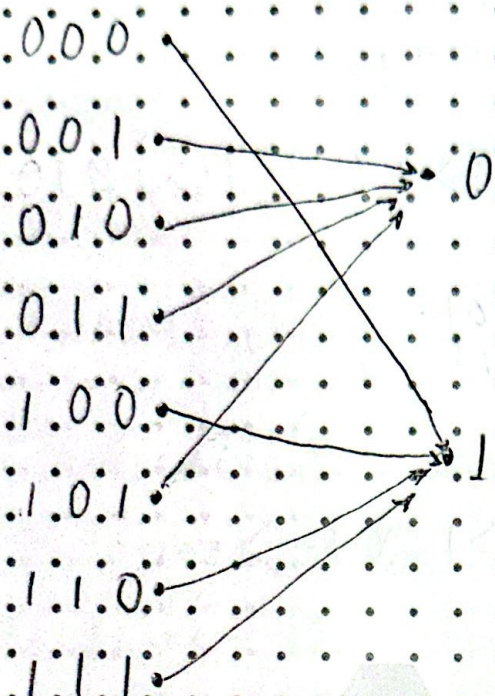
Solucion Ejercicio 10 Parcial 1



Estado Inicial

$$|\psi_0\rangle = |01\rangle = |0\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

nos quedamos con la representación en base computacional seguimos con el siguiente estado.

$$|\psi_1\rangle = (H \otimes H^{\otimes n})|01\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |-\rangle$$

$$|\psi_2\rangle = U_f(H \otimes H^{\otimes n})(|01\rangle)$$

Según la función

Aplicación Oráculo - Phase Kickback

$U_f |x, y\rangle = |x, y \oplus f(x)\rangle$

$U_f(|x\rangle, |-\rangle) = (-1)^{f(x)} |x\rangle, |-\rangle$

$|\psi_2\rangle = \frac{1}{\sqrt{2}} \left((-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle \right) \otimes |-\rangle$

$f(x) = f(x_i)$ $\quad f(0) = 0$

$\quad\quad\quad\quad\quad\quad\quad\quad f(1) = 1$

Sustitución

$|\psi_2\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \otimes |-\rangle$

$|\psi_3\rangle = (H \otimes I) U_f (H \otimes H^{\otimes n}) (|01\rangle)$

$|\psi_3\rangle = |1\rangle \otimes |-\rangle$ ya que sabemos

$H(|0\rangle - |1\rangle) = |1\rangle \sqrt{2}$

$|\psi_3\rangle = |1\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$

$|\psi_3\rangle = \frac{1}{\sqrt{2}} (|10\rangle - |11\rangle)$

Amplitudes	Estado	Amplitud
	00	0
	01	0
	10	$\frac{1}{\sqrt{2}}$
	11	$-\frac{1}{\sqrt{2}}$

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PROBABILIDADES

$P(|10\rangle) = \frac{1}{2}$

$P(|11\rangle) = \frac{1}{2}$

$P(|00\rangle) = 0$

Clasificación función

Si primer Qubit = 0 → función Constante

Si primer Qubit = 1 → función Balanceada

Obtenemos $|1\rangle$ función balanceada

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